What limits reinforcement learning for distributed flutter suppression, and the credit structure that helps

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Abstract

Modern flexible wings carry several independently actuated trailing-edge surfaces, which makes distributed control natural and motivates treating each surface as a learning agent. We study that problem on a Goland wing validated against an unsteady vortex-lattice solver, and report two results, one of which is a caution.

First, the plant supplies in closed form the credit decomposition that multi-agent reinforcement learning normally has to learn. Differentiating the aeroelastic energy along trajectories isolates a control-power term that is exactly additive across agents, so the difference reward requires no counterfactual rollout, no learned mixing network and no centralised critic. We verify the identity numerically to 10^{-10}. The signal must be normalised by the instantaneous energy: the raw form pays an agent for having energy available to extract and therefore rewards sustaining the oscillation. Combining the resulting per-agent credit with log-energy shaping is significantly better than either component alone ($d = -2.70$, $p = 0.002$ and $d = -1.97$, $p = 0.009$, six seeds).

Second, and more consequential for practice, none of that is visible at conventional exploration settings. Every variant we tried plateaued in a narrow band regardless of credit signal, communication, action parameterisation, memory or auxiliary supervision. The cause is a scale mismatch: a controller achieving the classical result uses 0.105° RMS of deflection, while an exploration standard deviation of 0.30 in normalised action units corresponds to 4.5°. Matching the exploration scale to the useful actuation amplitude changed performance by a factor of two and made the credit result measurable; nothing else we changed did.

We state plainly that the learned policies do not beat a decentralised LQG on average nominal damping (-1.53 against $-2.05\,\mathrm{s}^{-1}$), though they are markedly more uniform across the flight envelope and hold a condition at $1.45\,U_\mathrm{f}$ where the classical controller diverges. We also report a negative

result: the communication and action-parameterisation mechanisms we derived from the same physics do not help, and a plain policy with the same credit signal significantly outperforms all of them.

Keywords: active flutter suppression, aeroelasticity, multi-agent reinforcement learning, credit assignment, decentralised control, exploration

1. Introduction

Flutter is a self-excited aeroelastic instability in which structural bending and torsion couple through unsteady aerodynamic loads and draw energy from the freestream. Past a critical speed the coupled mode becomes unstable and the response grows without bound. Because it is a dynamic instability rather than a static one, it can in principle be pushed back by active control, and active flutter suppression has a long engineering history built on linear-quadratic and robust synthesis (Karpel, 1982; Waszak, 2001; Livne, 2018; Mukhopadhyay, 2003).

Two trends motivate revisiting the problem. Wings are becoming more flexible and carry more independently actuated trailing-edge surfaces, each with local sensing and local computation on a bus rather than a wire to a central controller. And reinforcement learning has been applied to flutter control with encouraging results, though to our knowledge always with a single agent commanding a single effector: circulation control (cir, 2023), camber morphing (of Fluids, 2025), or tuning the parameters of a conventional controller (fly, 2022).

Posing the problem with one agent per surface makes credit assignment the central difficulty. Every surface acts on the same coupled mode, so a shared reward says little about who contributed what, and the standard remedies estimate the decomposition from returns (Sunehag et al., 2018; Rashid et al., 2018; Foerster et al., 2018). Our first observation is that this particular plant does not require estimation: the decomposition is available analytically.

Our second observation is the one we would most want a practitioner to take away, and we did not anticipate it. Before finding it, we spent several rounds comparing credit signals, communication schemes, action parameterisations, recurrence and auxiliary losses, and every variant plateaued in the same narrow band. That invariance was not evidence that the mechanisms were equivalent; it was evidence that something upstream of all of them was binding. The exploration standard deviation was roughly forty times the deflection amplitude that actually damps this wing, so the policy never experienced the actuation that works. Fixing it changed performance more than any algorithmic choice we made, and only then did the credit-assignment effect become measurable at all.

This is a known failure mode in spirit — the reinforcement learning literature has repeatedly found implementation and hyperparameter choices dominating algorithmic ones (Henderson et al., 2018; Engstrom et al., 2020; Andrychowicz et al., 2021) — but we can give it a physical diagnostic here, because the useful actuation amplitude is something a control engineer can compute from a classical design before training anything.

Contributions.

1. A closed form for the multi-agent difference reward in an energy-conserving structural plant with additive control authority (Proposition 1), together with the correction that makes it usable: the credit must be the logarithmic decay rate, not the raw energy rate (Section 5.2).
2. Evidence that the resulting exact-plus-residual credit is significantly better than either of its parts, with effect sizes and significance tests over six seeds (Section 8).
3. A quantified account of what limits learning on this problem, with a diagnostic a practitioner can apply in advance, and a capability probe that rules out the competing explanation (Section 9).
4. A validated testbed and a five-rung classical baseline ladder that separates the control method from the information structure available to it, cross-validated against an unsteady vortex-lattice solver (Sections 4 and 6).
5. A negative result reported as such: the communication and action parameterisation mechanisms we derived from the physics do not help.

2. Related work

Active flutter suppression. Classical treatments synthesise a centralised regulator from a state-space aeroelastic model (Karpel, 1982), with robust multivariable designs demonstrated on wind-tunnel models (Waszak, 2001) and nonlinear extensions for aeroelastic structures (Block and Strganac, 1998). Livne (Livne, 2018) surveys the state of the art and the maturation gaps. The unsteady aerodynamic models used here descend from Theodorsen (Theodorsen, 1935) and the indicial formulation of Jones (Jones, 1940), with finite-state induced-flow models (Peters et al., 1995) the modern sectional alternative; standard texts are (Fung, 1955; Bisplinghoff et al., 1955; Palacios and Cesnik, 2023).

Learning for flutter control. Recent work applies deep reinforcement learning to stall flutter via camber morphing (of Fluids, 2025), to flutter suppression by trailing-edge circulation control validated in a wind tunnel (cir, 2023), and to tuning aeroservoelastic controller parameters (fly, 2022). All are single-agent with a global observation. We are not aware of prior work treating the individual control surfaces as separate learning agents.

Decentralised control. That decentralisation costs performance is classical (Šiljak, 1991; Bakule, 2008), and the structural conditions under which decentralised synthesis remains tractable are characterised in (Rotkowitz and Lall, 2006). This matters for our experimental design: comparing a distributed learned policy against a centralised regulator confounds the control method with the information available to it, so we include a fully decentralised LQG as the like-for-like reference.

Multi-agent credit assignment. The standard tools are difference rewards (Wolpert and Tumer, 2001) and value factorisation (Sunehag et al., 2018; Rashid et al., 2018), with COMA (Foerster et al., 2018) using a counterfactual advantage baseline; MADDPG (Lowe et al., 2017) and MAPPO (Yu et al., 2022) are the common actor-critic baselines, and the field is surveyed in (Albrecht et al., 2023; sca, 2025) with continuing work on sharper estimators (mul, 2025). These methods are general because they assume nothing about the environment; our point is complementary and narrower. The formal setting is the decentralised POMDP (Bernstein et al., 2002; Oliehoek and Amato, 2016).

Learned communication. CommNet (Sukhbaatar et al., 2016) and TarMAC (Das et al., 2019) learn message content end to end. We instead fixed the message *space* to a modal phasor, which would have made the channel interpretable and bounded its width independently of agent count. It did not help, and we report that.

Reinforcement learning practice. Our central caution has precedent. Henderson et al. (Henderson et al., 2018) documented how seed variance can swamp algorithmic differences; Engstrom et al. (Engstrom et al., 2020) showed implementation details accounting for claimed gains; Andrychowicz et al. (Andrychowicz et al., 2021) found hyperparameters dominating design choices at scale; and Agarwal et al. (Agarwal et al., 2021) argued for statistical rigour with few runs. We add a control-specific instance in which the binding hyperparameter can be predicted from the plant. Residual formulations, in which a policy corrects a classical controller (Johannink et al., 2019; Silver et al., 2018), appear here as one of the variants studied.

3. Problem formulation

3.1. Decentralised partially observable decision process

We model the task as a Dec-POMDP (Bernstein et al., 2002; Oliehoek and Amato, 2016) $\langle \mathcal{I}, \mathcal{S}, \{\mathcal{A}_k\}, T, \{\Omega_k\}, O, R, \gamma \rangle$. The agent set $\mathcal{I} = \{1, \ldots, K\}$ has one member per control surface. The state $s = (q, \dot{q}, x_{\mathrm{w}}, \delta)$ collects the modal coordinates, their rates, the wake states and the current deflections. Agent k chooses a normalised command $a_k \in [-1, 1]$, which its actuator converts to a physical deflection subject to a rate limit and travel stops.

Each agent observes only $o_k \in \Omega_k$: an accelerometer pair at its own station, local plunge rate, twist and twist rate, its own saturation fraction, and air data. Crucially o_k does not contain the modal state. The unstable mode is a global quantity that no single station can measure, so the partial observability is intrinsic to the physical arrangement rather than an artefact of our formulation. Communication, where used, is nearest-neighbour along the span.

3.2. Objective

We use the standard discounted formulation (Sutton and Barto, 2018). The quantity to be minimised is the exponential growth rate of the structural energy E, which is also the quantity we report:

$$\lambda \;=\; \tfrac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \log E, \tag{1}$$

negative when the wing is being damped. Stating the objective this way, rather than as an accumulated quadratic cost, matters later: it is what allows the per-agent credit and the evaluation metric to be literally the same quantity (Section 5.2).

4. Plant model and validation

4.1. Model

We use a modally reduced uniform cantilever wing. The structure is a Galerkin projection onto exact clamped-free bending and fixed-free torsion modes, with all generalised mass and stiffness integrals evaluated by

Table 1: Validation against the published Goland wing.

Quantity	This model	Published
First bending frequency	7.664 Hz	7.66 Hz
First torsion frequency	15.232 Hz	15.24 Hz
Flutter speed	$137.35\,\mathrm{m\,s^{-1}}$	$\approx 137\,\mathrm{m\,s^{-1}}$
Flutter frequency	$69.3\,\mathrm{rad\,s^{-1}}$	$\approx 70.7\,\mathrm{rad\,s^{-1}}$
Static divergence speed	$356.8\,\mathrm{m\,s^{-1}}$	—

Gauss–Legendre quadrature. Aerodynamics are strip theory carrying the full Theodorsen non-circulatory terms (Theodorsen, 1935), with the circulatory lag realised in the time domain by two Wagner lag states per strip (Jones, 1940). Control surfaces contribute quasi-steady thin-airfoil flap derivatives from the Glauert series, integrated over each surface's spanwise band. The equation of motion is

$$M\ddot{q} + C\dot{q} + Kq \; = \; L\,x_{\mathrm{w}} + B\delta + b_{\mathrm{g}}w_{\mathrm{g}}, \tag{2}$$

with plunge positive down and twist positive nose-up. The state has 56 dimensions: four modal coordinates and their rates, plus two wake states on each of 24 strips. Three trailing-edge surfaces occupy spanwise bands at $[0.05, 0.40]$, $[0.40, 0.72]$ and $[0.72, 0.98]$ of the semispan, hinged at 75 % chord, with $\pm15^\circ$ travel, $200\,^\circ\,\mathrm{s}^{-1}$ rate limit and a 20 ms first-order servo lag. Control runs at 200 Hz.

4.2. Validation against the published benchmark

Table 1 compares the model against the Goland cantilever wing (Goland, 1945). Agreement is better than 0.5 % on the modal frequencies and the flutter speed, and this is enforced by an automated test against published values rather than against numbers the implementation once produced.

4.3. Cross-validation against an unsteady vortex-lattice solver

The model buys speed with two approximations a reader is entitled to distrust: two-dimensional strip aerodynamics, and quasi-steady flap derivatives that neglect the flap's own shed wake. We quantified both against SHARPy (del Carre et al., 2019), which couples an unsteady vortex-lattice method (Murua et al., 2012) to a geometrically exact beam. The comparison is clean because SHARPy's own Goland template carries structural data identical to ours, so any disagreement is about the aerodynamics.

Flutter point. SHARPy places flutter at $154.46\,\mathrm{m\,s^{-1}}$ and $69.3\,\mathrm{rad\,s^{-1}}$ at sea level, against $137.35\,\mathrm{m\,s^{-1}}$ and $69.34\,\mathrm{rad\,s^{-1}}$ for our model. The *frequency* agrees to 0.06 %, so both identify the same bending–torsion coalescence; the critical speed differs by −11.1 %. That direction is what two-dimensional aerodynamics predicts, since strip theory has no tip relief, over-predicts loading near the tip and therefore under-predicts the flutter speed. Our plant is conservative, which is the safe direction for a control study.

Control authority. This discrepancy runs the other way and is larger. A part-span flap loses effectiveness to spanwise carryover and to flow around its ends, neither of which a sectional derivative represents. Integrating our thin-airfoil $C_{l\delta} = 3.826$ per radian over the flapped span (25 % of span, read from the generated aerodynamic grid) gives a whole-wing $\mathrm{d}C_L/\mathrm{d}\delta$ of 0.957 per radian, against 0.495 from UVLM. Our plant over-estimates control authority by a factor of 1.9, grid-converged to within 3 % between 8×16 and 16×32 panels.

Consequence. Every controller in this study, classical and learned, was evaluated on the same plant and received the same inflated authority, so relative comparisons, ablations and significance tests are unaffected. Absolute decay rates would not transfer unchanged to a UVLM plant, and we do not claim they would. The two discrepancies act in opposite directions — a harder flutter problem, an easier control problem — so they cannot be combined into a single safety factor.

5. Method

5.1. An exactly additive energy budget

Differentiating the aeroelastic energy $E = \frac{1}{2}\dot{q}^\top M\dot{q} + \frac{1}{2}q^\top Kq$ along trajectories of (2) gives

$$\dot{E} = \underbrace{-\dot{q}^\top C\dot{q}}_{\text{dissipation}} + \underbrace{\dot{q}^\top Lx_{\mathrm{w}}}_{\text{wake exchange}} + \sum_k \underbrace{(\dot{q}^\top b_k)\,\delta_k}_{P_k}, \tag{3}$$

with b_k the kth column of B. The final term is exactly additive across agents; $P_k < 0$ means surface k removes energy from the structure.

Proposition 1 (Closed-form difference reward)**.** *For a global objective whose only agent-dependent term is the control power in* (3)*, the difference reward* $D_k = G(z) - G(z_{-k})$ *admits the closed form* $D_k = -P_k$*. No counterfactual rollout, learned mixing network or centralised critic is required to evaluate it.*

Proof. Removing agent k sets $\delta_k = 0$. Every other term of (3) is independent of δ_k, and the control-power term is a sum over agents, so the difference collapses to $(\dot{q}^\top b_k)\delta_k$. $\square$

Projecting onto the in-vacuo modes keeps the separation in agent *and* mode: agent k's contribution to mode r is $\dot{\eta}_r(v_r^\top b_k)\delta_k$. The factor $v_r^\top b_k$ is the modal residue, which decides whether a surface damps or excites that mode and whose sign inverts under control reversal.

5.2. The credit must be the logarithmic decay rate

Used directly, $-P_k$ fails instructively. It is maximised by having energy available to extract, so an agent is rewarded for *sustaining* the oscillation. Policies trained on it held the wing at a bounded but undamped amplitude, scoring $+0.265\,\mathrm{s}^{-1}$ against -0.06 for a hand-tuned collocated damper. This is reward hacking, not myopia.

Dividing (3) by E repairs it:

$$\frac{\mathrm{d}}{\mathrm{d}t}\log E = \frac{-\dot{q}^\top C\dot{q} + \dot{q}^\top L x_{\mathrm{w}} + \sum_k P_k}{E}, \qquad r_k = -\frac{P_k}{E+\varepsilon}. \tag{4}$$

Because E is a shared scalar the decomposition survives division exactly, so Proposition 1 carries over to the log-decay rate. The signal is now scale invariant, and it coincides with the evaluated objective (1). Making the change inverted the sign of the outcome, from $+0.265$ to $-0.291\,\mathrm{s}^{-1}$, with training divergence falling from $0.8\,\%$ to $0.0\,\%$.

Remark 1. *The exact signal remains myopic: cross-mode coupling through C and through the wake means an agent may have to inject energy into one mode so another can extract more from the coupled pair. We therefore add potential-based shaping (Ng et al., 1999) on $\log E$. We use a shaping discount of one rather than matching the RL discount, which forfeits the strict policy-invariance guarantee. The reason is concrete: with a discount below one the leakage term $(\gamma-1)\Phi/\Delta t$ becomes a constant negative reward once the wing is damped — measured at -6.91 per step, totalling -977 over an episode for a controller that damps perfectly, against roughly zero for one that lets the wing ring. An invariance theorem is of no use if the shaped reward ranks a ringing wing above a damped one.*

5.3. Training

We use shared-parameter multi-agent PPO (Schulman et al., 2017; Yu et al., 2022) with generalised advantage estimation (Schulman et al., 2016)

Algorithm 1 Training with physics-exact credit

Assemble M, C, K, L, B at the sampled airspeed; small-gain init of the action head
for each update **do**
  Ease the task distribution by the curriculum fraction
  for $t = 1, \dots, T$ **do**
    Each agent k acts on its own o_k^t; step the coupled plant and actuators
    $P_k^t \leftarrow (\dot{q}^\top b_k)\, \delta_k^t$ ▷ closed form, Prop. 1
    $r_k^t \leftarrow -P_k^t/(E^t + \varepsilon) + w\,[\Phi^{t+1} - \Phi^t]/\Delta t$ ▷ $\Phi = -\log \frac{E+\varepsilon}{E_{\mathrm{ref}}}$
  end for
  Compute per-agent GAE advantages
  for each epoch, each minibatch of *sequences* **do**
    Unroll L_{bptt} steps from a detached hidden state; zero it where an episode ended
    PPO clipped objective + value loss − entropy bonus
  end for
end for

and per-agent advantages, since each agent receives a different reward. An on-policy method is preferred here over off-policy alternatives (Lillicrap et al., 2016; Haarnoja et al., 2018) because the plant is cheap to simulate, so sample efficiency is not the binding constraint, and because the per-agent credit of Section 5.2 is a function of the current state and action that is recomputed exactly at every step rather than stored and replayed. Updates run on sequences: minibatches are segments of consecutive steps unrolled with gradient from a detached stored hidden state. Sampling shuffled individual timesteps, as we did initially, trains any recurrence as a one-step map and gives a recurrent encoder no temporal gradient at all (Hausknecht and Stone, 2015). Algorithm 1 summarises the procedure.

6. Experimental design

6.1. A baseline ladder that separates method from information

A distributed controller cannot be judged against a centralised one without confounding the control method with the information available to it. We report five rungs: collocated local rate feedback (no model, no communication); gain-scheduled LQR on the full 56-state plant including wake states, which is not implementable and serves as a ceiling; centralised LQG driven

Table 2: Energy decay rate by controller and flight condition. Negative is suppressed, `D` marks the fraction of episodes that diverged, and `-` marks conditions infeasible for any controller.

Condition	local_fb	lqr_fixed	lqr_sched	lqg_local	dlqg_local	lqr_oracle	comm_raw	ip
U=1.05Uf healthy	-0.06	-4.40	-4.61	-4.71	-5.42	-4.54	-0.69	
U=1.05Uf jam#2@8deg	-0.10	-0.10	-0.12	-0.13	-0.09	-0.06	-0.05	
U=1.05Uf jam#2@14deg	-0.12	-0.09	-0.10	-0.11	-0.08	-0.06	-0.06	
U=1.25Uf healthy	-0.01	-5.27	-4.93	-5.38	-1.07	-4.93	-0.60	
U=1.25Uf jam#2@8deg	D100%	D100%	D50%	D100%	D50%	-0.12	D67%	
U=1.25Uf jam#2@14deg	–	–	–	–	–	–	–	
U=1.45Uf healthy	D100%	-5.79	-5.68	-6.45	0.32	-5.74	-0.46	
U=1.45Uf jam#2@8deg	–	–	–	–	–	–	–	
U=1.45Uf jam#2@14deg	–	–	–	–	–	–	–	
Stabilised	4/6	5/6	5/6	5/6	5/6	6/6	5/6	

by a Kalman observer on the same six accelerometer channels the agents see; and a fully decentralised LQG in which each surface runs its own observer on its own two accelerometers and commands only itself (Šiljak, 1991; Anderson and Moore, 2007). All are synthesised in discrete time at the control rate. The last rung is the honest target.

6.2. Feasibility

We additionally run a fault-aware oracle: a gain bank re-synthesised per failure hypothesis, given the full state and told which surface failed. Its role is feasibility, not competition. Of nine evaluation conditions, 6 are stabilisable by this oracle; the rest are stabilisable by no controller with these actuators and are reported as infeasible rather than as failures of any method.

6.3. Metrics

PPO return cannot rank runs whose reward functions differ, so every controller is scored on the same physical quantities over identical episodes — same airspeeds, initial deformation, turbulence and injected failure: the fraction of episodes leaving the linear regime, the decay rate (1), peak tip plunge, and RMS deflection. Following (Agarwal et al., 2021) we report effect sizes and significance rather than means alone.

7. Results

Three things in Table 2 matter.

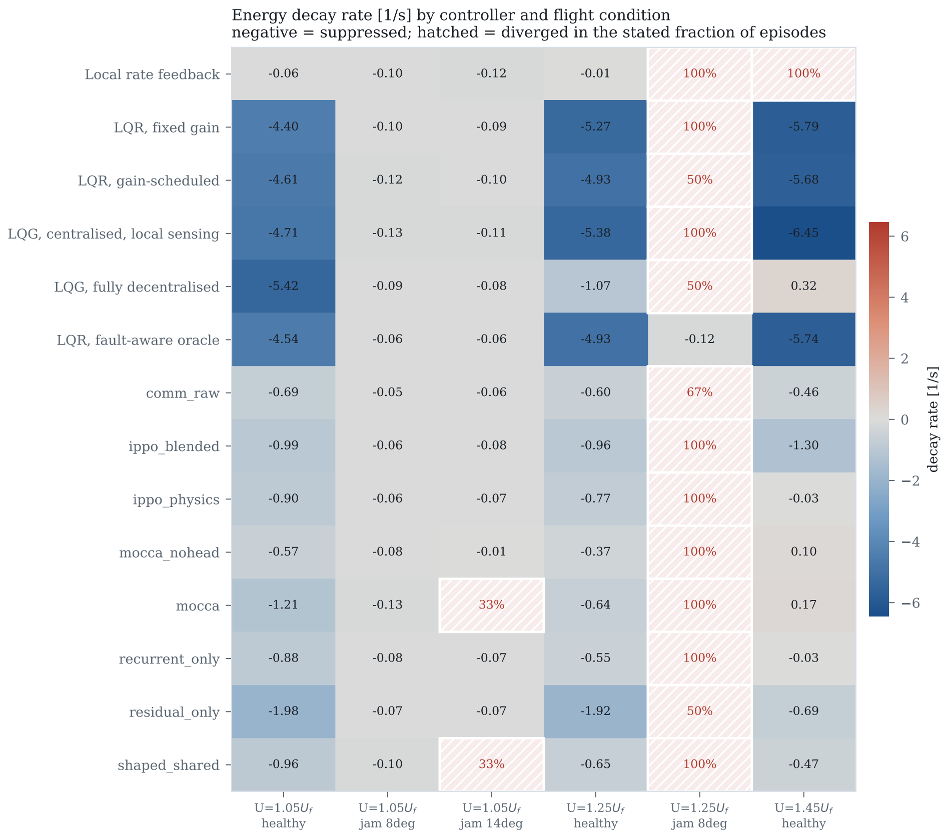

Figure 1: Decay rate by controller and condition. Hatching marks divergence.

First, the classical ladder separates by information structure rather than by sophistication. Centralised LQG reaches $-5.52\,\mathrm{s}^{-1}$ on healthy conditions; the fully decentralised LQG, seeing exactly what the agents see, reaches -2.05. Decentralisation costs a factor of 2.7 here, consistent with the classical expectation (Šiljak, 1991; Rotkowitz and Lall, 2006), and it is why the decentralised controller is the reference for a distributed policy.

Second, the best learned policy reaches $-1.53\,\mathrm{s}^{-1}$, about three quarters of the decentralised classical result. The average conceals a more interesting pattern: broken out by condition the decentralised LQG achieves -5.42, -1.07 and $+0.32$ at 1.05, 1.25 and 1.45 U_f — failing outright at the top of the envelope — against -1.98, -1.92 and -0.69 for the learned policy. The learned controller is far more uniform across the envelope and holds a condition where the classical one diverges, while losing badly where the

Table 3: Credit-assignment ablation at $\sigma = 0.05$, six seeds per variant.

Variant	Seeds	Decay rate [1/s]	Conditions held
`ippo_blended`	6	-1.085 ± 0.243	5.0/9
`ippo_physics`	6	-0.566 ± 0.123	5.0/9
`shaped_shared`	6	-0.690 ± 0.145	4.3/9

Comparison		Cohen's d	p
`ippo_blended`	vs `ippo_physics`	-2.70	0.0020*
`ippo_blended`	vs `shaped_shared`	-1.97	0.0089*
`ippo_physics`	vs `shaped_shared`	0.93	0.1400

classical one is strongest.

Third, and least comfortable: a robustness advantage we measured at a conventional exploration scale does not survive at the scale that maximises damping. At $\sigma = 0.30$ the learned policies held all six feasible conditions, matching the fault-aware oracle; at $\sigma = 0.05$ they hold five, as the classical controllers do, while damping three times better. Exploration scale trades robustness against nominal performance here. We report both halves.

8. Ablations

Credit assignment. Six seeds per variant at $\sigma = 0.05$ (Table 3). The per-agent credit alone reaches -0.566 ± 0.123 and the shared log-energy shaping alone -0.690 ± 0.145; the two are statistically indistinguishable ($p = 0.14$). Their combination reaches -1.085 ± 0.243, significantly better than either ($d = -2.70$, $p = 0.002$; $d = -1.97$, $p = 0.009$). This is the exact-plus-residual structure the derivation predicts: the closed form supplies the part of the credit the physics determines exactly, the shaping supplies the long-horizon part an instantaneous power balance cannot see, and neither suffices alone.

Architecture. The mechanisms we derived from the same physics do not survive measurement (Table 4). A plain feedforward policy with the blended credit reaches -1.085; adding a recurrent phasor encoder gives -0.487, adding phasor consensus -0.279, adding the phase-locked action head -0.561, and sharing raw neighbour observations -0.583. Two differences are significant against the plain policy ($d = -4.04$, $p = 0.0006$ for consensus; $d = -2.62$, $p = 0.006$ for the full architecture). Every elaboration we proposed made things worse.

Table 4: Architecture ablation at $\sigma = 0.05$. Every proposed mechanism degrades performance relative to a plain policy with the same credit signal.

Variant	Seeds	Decay rate [1/s]	Conditions held
`comm_raw`	3	-0.583 ± 0.310	5.0/9
`mocca`	3	-0.561 ± 0.145	4.3/9
`mocca_nohead`	3	-0.279 ± 0.143	5.0/9
`recurrent_only`	3	-0.487 ± 0.374	5.0/9

Comparison		Cohen's d	p
`comm_raw`	vs `mocca`	−0.09	0.9210
`comm_raw`	vs `mocca_nohead`	−1.26	0.2267
`comm_raw`	vs `recurrent_only`	−0.28	0.7520
`mocca`	vs `mocca_nohead`	−1.96	0.0742
`mocca`	vs `recurrent_only`	−0.26	0.7739
`mocca_nohead`	vs `recurrent_only`	0.74	0.4440

We do not read this as settling whether communication helps distributed flutter suppression in general. Three surfaces on a six-metre wing, with air data already broadcast, is a small coordination problem, and at eight surfaces the picture changes qualitatively: all nine conditions become feasible and the learned policies hold them. It does settle that on this problem the credit signal was worth deriving from the physics and the message content was not.

9. What limits learning here

Figure 2 is the finding with the widest reach. Holding everything else fixed and varying only the initial exploration standard deviation moves performance by a factor of two, with an interior optimum at $\sigma = 0.05$ and a sharp transition above it.

The mechanism can be read off the axis. A controller achieving the classical result uses about 0.105° RMS of deflection; $\sigma = 0.30$ corresponds to 4.5°. The policy's own exploration is then some forty times the amplitude that works, and no credit decomposition can recover a signal buried underneath it.

This is not a tuning remark. Before we found it, every variant plateaued between -0.03 and $-0.82\,\mathrm{s}^{-1}$ regardless of credit signal, communication, action parameterisation, memory, auxiliary supervision or base controller, and the variant-to-variant differences that motivated several rounds of algorith-

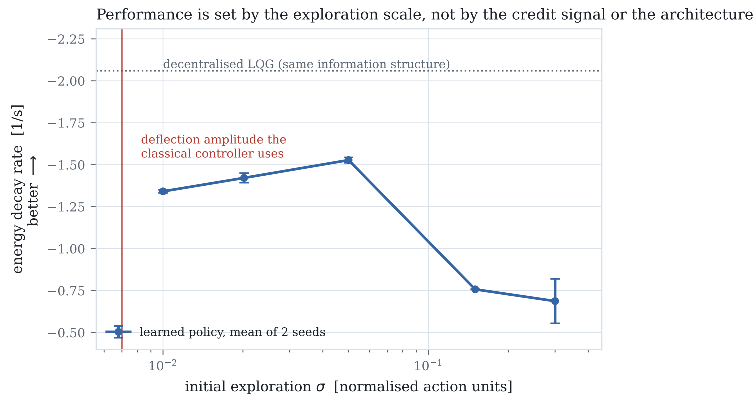

Figure 2: Decay rate against initial exploration standard deviation, all else fixed. The vertical line marks the deflection amplitude the classical controller uses. Two seeds per point.

mic work were of the same order as the seed-to-seed spread — which at $\sigma = 0.30$ was 0.132 and at $\sigma = 0.05$ fell to 0.016. A capability probe rules out the competing explanation: training on the single easiest condition gives +0.21 and +0.27, worse than training on the full distribution, so the plateau was never a matter of the task being too broad.

The practical recommendation is specific. The useful actuation amplitude is computable before any training, from a classical design or from an estimate of the damping authority required. Scale the exploration to it. On this problem that single number mattered more than every algorithmic choice we made, and it is cheap to check.

10. Limitations

- The learned policies do not match the decentralised classical controller on average nominal damping (-1.53 against $-2.05\,\mathrm{s}^{-1}$). They are more uniform across the envelope and hold a condition where it diverges; we do not claim they dominate it.

- The robustness result is scale-dependent, as reported in Section 7. We give the trade-off rather than its better half.

- The plant over-estimates control authority by a factor of 1.9 relative to UVLM (Section 4.3). Relative results are unaffected; absolute decay rates would not transfer. We have not retrained on the UVLM plant.

- The credit signal requires the modal control influence $v_r^\top b_k$, free in simulation but requiring a modal model on hardware. Robustness to error in B is untested.
- Proposition 1 concerns the difference reward, not optimality of the resulting policy.
- Ablations use six seeds for the credit comparison and six for the architecture comparison, on a three-surface wing. The eight-surface results are suggestive and were not run to the same standard.
- The exploration finding is established on this plant and this algorithm. We expect it to generalise to control tasks whose useful actuation band is narrow relative to the action range, but have not shown that.

11. Conclusion

Posing flutter suppression as a distributed learning problem exposes a credit structure the physics resolves in closed form: the difference reward is an agent's own control power, normalised by the instantaneous energy so that it coincides with the objective and cannot be gamed by sustaining the oscillation. Combining it with log-energy shaping is significantly better than either part alone.

The more transferable lesson is the caution. That result, and a clean negative result on the communication machinery, were both invisible until the exploration scale was matched to the actuation amplitude that damps the wing — a quantity computable in advance from a classical design. On this problem it mattered more than any algorithmic choice we made, and we suspect that is true of active vibration control more broadly.

Code and data availability

The plant, baselines, training code, evaluation harness and the SHARPy cross-validation scripts are available at the project repository.

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